EMERGENCY DEPARTMENT ASSESSMENT OF CHEST PAIN AND SUSPECTED ACUTE CORONARY SYNDROME

AlfredHealth

Target audience

This guideline has been developed to assist clinicians caring for patients presenting to the Alfred Emergency and Trauma Centre or Sandringham Hospital Emergency Department with Chest pain (or chest pain-equivalent symptoms) where Acute Coronary Syndrome (ACS) is suspected.

Purpose

To encourage a standardised and safe approach to the management of patients with suspected ACS, by providing an evidence-based framework for the assessment, management and disposition of these patients, together with recommendations regarding outpatient investigations and follow up where these are appropriate.

Patient Selection

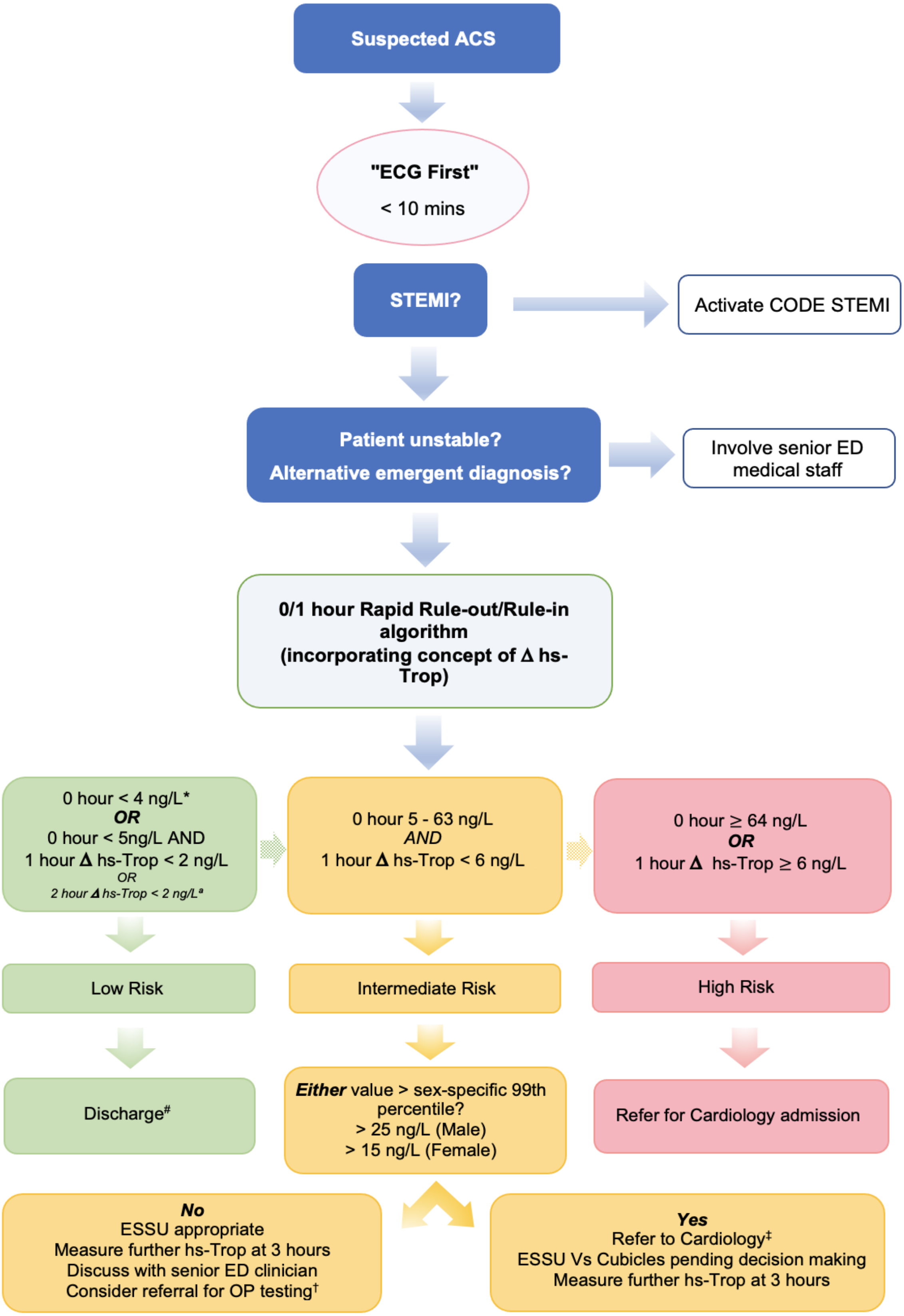
Patients presenting with suspected ACS comprise a diverse group of patients. Broadly they include any patient presenting with chest pain or a chest pain-equivalent symptom that is suspicious for ACS.

The concept of **chest pain equivalent symptoms** acknowledges that not all patients presenting with ACS will have typical cardiac-sounding chest pain. Such symptoms can include dyspnoea, diaphoresis, left or right arm pain, shoulder pain, neck or jaw pain, upper back pain, and epigastric pain. Identifying ACS presenting with such symptoms can be extremely challenging and requires a high level of vigilance on the part of both streaming nurses and initial treating clinicians.

Summary

Some patients presenting with chest pain will have life-threatening and time-critical diagnoses. The steps in this guideline are therefore ordered in a sequence according to clinical priorities which must be followed from the beginning as a linear workflow (see Fig.1).

Figure 1. Suspected ACS workflow



*** If presentation is >2 hours from symptom onset, patients can be designated ruled-out on the basis of a single high sensitivity troponin (hs-Trop) <4 ng/L. Patients presenting <2 hours from symptom onset require serial troponins with at least one troponin >2 hours from symptom onset.**

[a] Whether a 1 hour or 2 hour serial hs-Trop is taken for patients with a 0 hour hs-Trop <5 ng/L will be determined by time since symptom onset at presentation. Patients in this category presenting >60 minutes from symptom onset will require a 1 hour hs-Trop. Patients presenting <60 minutes from symptom onset in this category require a 2 hour hs-Trop (i.e. 2 hours from 0 hour trop). This is to enable clear use of assay-specific cut-off values.

[#] It is recommended that junior medical staff calculate and document HEART Score prior to discharge (see Appendix). Patients with a heart score ≥ 4 should be discussed with either a senior ED registrar or consultant prior to discharge. These patients may, in some cases, require referral to cardiology outpatients. Low risk patients with a HEART score <4 can be safely discharged to care of GP.

It should be noted that use of the hs-Trop 0/1 hour algorithm alone, absent consideration of risk scores such as HEART, has been shown to be non-inferior to strategies that include such scores, with the additional advantage of decreasing resource use and unnecessary testing.

[†]Patients in this category with a 3 hour hs-Trop < sex-specific 99th percentile and a HEART Score ≤ 3 can usually be discharged with no further follow up. Higher risk patients (as determined by the senior ED clinician) can be discussed with cardiology and may be either admitted or referred for early outpatient testing.

[‡]These patients will usually either be admitted and treated for ACS or admitted for inpatient testing (even if ACS diagnosis unclear). Occasionally - and only after in-person cardiology registrar review - patients in this group may be discharged with a plan for outpatient follow up.

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GUIDELINE

1. CLINICAL WORKFLOW

1.1 Step 1

1.1.1. Stream patient to a *monitored cubicle* (ATS triage category will usually be 2 or higher, though this is at the discretion of the streaming nurse)

1.1.2. **"ECG First"** - Capture and link to Cerner a 12 lead ECG **within 10 minutes** of patient arrival.

NOTE. This should occur in all suspected ACS **prior to** obtaining other vital signs, and **prior to** completion of the remainder of the initial nursing assessment.

1.1.3. Have the linked ECG sighted immediately by the treating doctor, or in their absence, by the admitting officer (AO).

Stop point

Is the patient having a STEMI?

YES → follow CODE STEMI guideline

NO → continue to step 2

1.2 Step 2

1.2.1 Complete initial nursing assessment including full set of vital signs.

1.2.2 Complete initial medical assessment with particular focus on

- **Consideration of other non-ACS emergent diagnoses** including pulmonary embolism (PE), aortic dissection and tension pneumothorax.
- Collection of **key historical data** including time of symptom onset, presence of associated features with a proven positive likelihood ratio (exertional pain, radiation to either arm, diaphoresis, vomiting with pain, and similarity to previous MI) and classical risk factors for coronary artery disease (smoking status, diagnosed hypertension, diabetes mellitus, hypercholesterolaemia and family history of premature coronary artery disease).
- Be vigilant to the greater risk of delayed or missed diagnosis of ACS in **women**, and to the higher risk of ACS in younger **first nations people** (always facilitate access to first nations liaison officers when desired and available)

1.2.3 give **aspirin** 300 mg, unless contraindicated

1.2.4 Offer **analgesia** according to symptoms

1.2.5 Chest X-ray (CXR) where indicated to exclude differential diagnoses

1.2.6 **Obtain a repeat ECG** for all patients 15 to 30 minutes after the first

NOTE: this will usually be prior to the patient being moved to another area

Stop point

Is the patient **unstable**?

Do they have abnormal vital signs: HR <60 or >110, SBP <90 mmHg, RR >30 or <10, $Sp0_2$ <90%, GCS <14)

Are there **concerning ECG changes**?

Including potentially ischaemic dynamic changes, other patterns concerning for ischaemia (e.g. widespread ST depression, Wellens syndrome), unstable arrhythmias, or high grade AV block.

Is there a more likely **emergent non-ACS diagnosis**?

PE? Aortic dissection? Tension pneumothorax? Acute abdomen?

YES involve senior ED clinician. Do not proceed to step 3

NO proceed to step 3

1.3 Step 3

1.3.1 Measure hs-Trop and follow the 0/1 hour Rapid Rule-out / Rule-in algorithm (see figure 2 or 3)

High sensitivity troponin (hs-Trop)

High sensitivity troponin assays, by definition, allow for the detection of troponin in 50-95% of healthy individuals. A hs-Trop level >99th percentile for healthy individuals is considered 'elevated' and, in the context of a consistent clinical presentation, is required for the diagnosis of myocardial infarction (MI) according to the fourth universal definition of MI.

Alfred pathology operates a high sensitivity Troponin I assay (Abbott Alinity) with the following sex-specific reference ranges:

Male	>25 ng/L	=	elevated
Female	>15 ng/L	=	elevated

Note on transgender patients

Currently the effects of exogenous supplemental sex hormones in transgender men and women on hs-Trop reference ranges is not clear. For safety, it is recommended to use the *lower* (female) reference range for all such patients until further evidence is available.

The 0/1 hour Rapid Rule-out / Rule-in algorithm

Diagnostic algorithms based around the measurement of high-sensitivity troponin (hs-Trop), and which classify patients into 'rule-out' (low risk), 'rule-in' (high risk) and intermediate (observation) groups, are now broadly recommended as standard of care for the emergency department assessment of suspected ACS[1, 2].

Such algorithms have been shown to safely identify patients that can be discharged from the ED ('low risk' patients - translating to a 30-day risk of MACE <1%) while reducing time to diagnosis, unnecessary testing and ED length of stay (LOS).

One such algorithm - the 0/1 hour algorithm - has been widely validated and is recommended by both the European Society of Cardiology (ESC)[2] and the Cardiac Society of Australia and New Zealand (CSANZ), as detailed in their most recent guideline (Diagnosis and Management of Acute Coronary Syndromes, 2025[1]).

Correct application of the 0/1 hour Rule-in / Rule-out algorithm:

- hs-Trop values are assay-specific. At the Alfred Hospital Emergency and Trauma Centre (ET+C) and during laboratory hours at Sandringham Hospital the **Abbott Alinity** hs-Trop analyser is used. For all patients at the Alfred, therefore, and at Sandringham during laboratory hours care is to be based on the hs-Trop values as included in figure 2 (primary algorithm).

- out of laboratory hours, from 2025, the **Quidel TriageTrue** point-of-care (POC) hs-Trop assay will be available in Sandringham Hospital and can be used as an alternative to the Abbott troponin analyser (figure 3, out-of-hours POC 0/1 hour algorithm).

It is vital to recognise that values from the two assays **cannot be used 'in combination'** for any one patient as this will lead to invalid conclusions about Δ hs-Trop and thresholds for rule-in/rule-out. When following the algorithms **one or other (never a combination)** of either the lab or POC values must be used to make decisions about any given patient.

Step 3 continued

Other important practice points:

1.3.2 **Unstable angina**. The number of patients diagnosed with unstable angina has decreased since the advent of hs-Troponin assays as many such patients have likely been reclassified as NSTEMI.

If true unstable angina is suspected (for example, in patients with ongoing typically cardiac-sounding pain, or those with crescendo pattern symptoms whereby chest pain or other ACS symptoms are occurring with less and less exertion over time) a senior ED clinician should be involved in decision making. Such patients are not suitable for the rapid Rule-in / Rule-out approach and may require discussion with cardiology.

1.3.3 In any patient with **recurrent or escalating symptoms**, consideration should be given to performing an additional hs-Trop. Generally, this additional sample should be taken 2 hours after the onset of recurrent symptoms, or 2 hours from maximal symptoms. These patients should be discussed with a senior ED registrar or consultant before additional any samples are drawn.

1.3.4 Repeat 12 lead ECGs must be recorded and linked within Cerner whenever hs-Trop samples are drawn or whenever a patient's pain changes or increases. **ACS cannot be ruled out in the presence of dynamic ECG changes** that are potentially consistent with ischaemia.

1.3.5 Where there is suspicion for pulmonary embolism (PE) or aortic dissection, **D dimer** should be measured for low risk patients.

Figure 2. Primary 0/1 hour rapid rule out / rule-in algorithm:

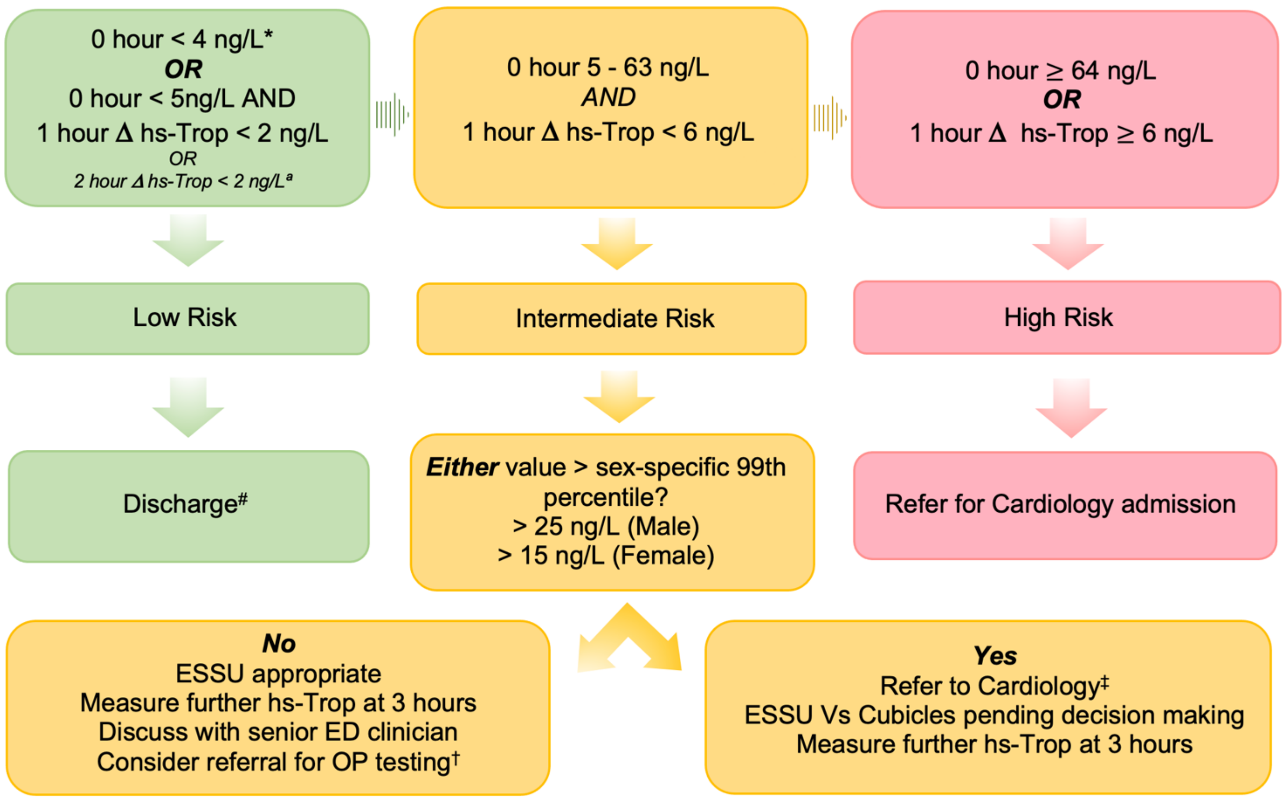

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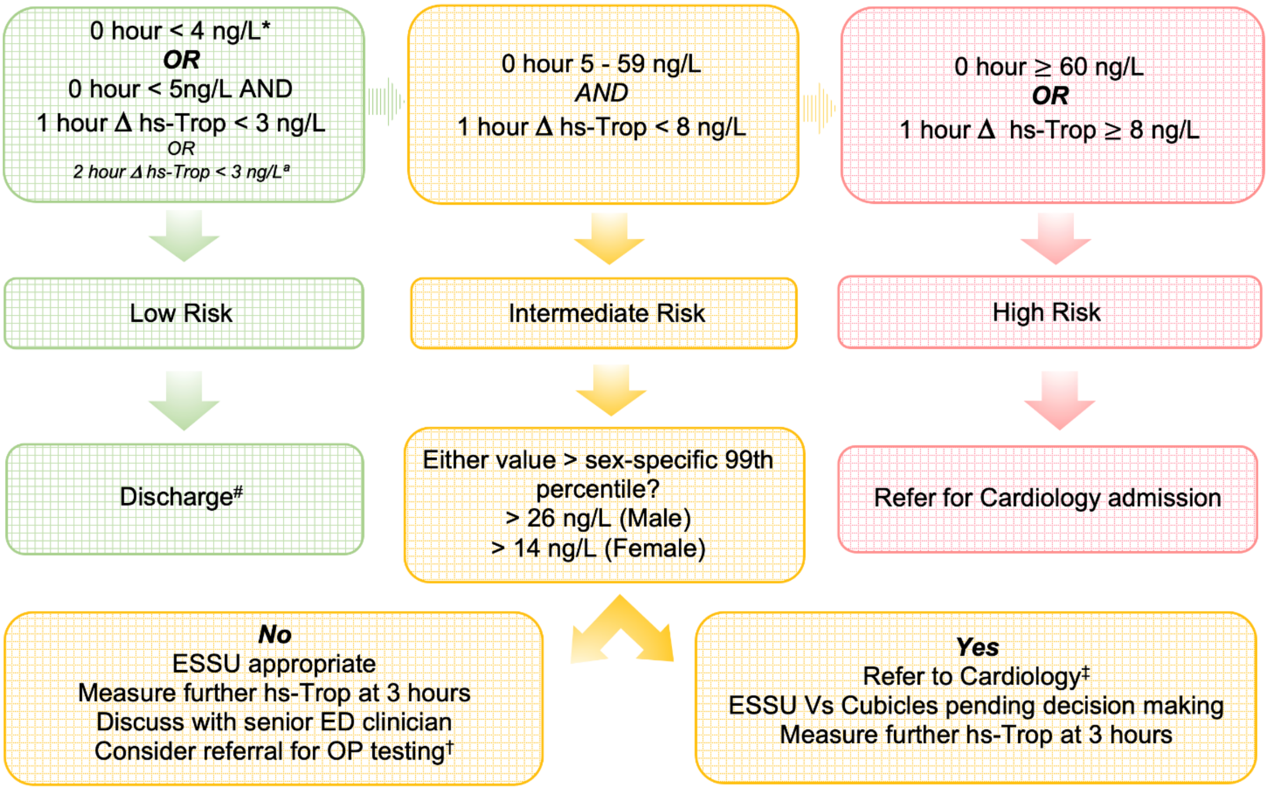
Figure 3. Out-of-hours (POC) pathway:



* see figure 1
# see figure 1
† see figure 1
‡ see figure 1

1.4 Step 4

Final disposition

1.4.1 Action patient disposition according to final risk status (low, intermediate or high)

Early outpatient follow up for intermediate risk patients

If following discussion with the cardiology unit or an ED consultant it is deemed appropriate to arrange outpatient follow up for an intermediate risk patient the following is recommended:

- Intermediate risk patients being discharged from the ED should, as a rule, be referred for early (within days of discharge) further testing for ischaemic heart disease and be followed up by a cardiologist.

- Patients being referred for early outpatient testing must be discussed with the cardiology registrar prior to discharge from the ED.

- Where possible, ED should facilitate ordering an appropriate outpatient test (*usually dobutamine stress echo or CT coronary angiography*) such that this occurs in parallel with any planned early cardiology review.

- The process for arranging these tests is detailed in box 1. The choice of test for any particular patient is best made in consultation with cardiology or a senior ED clinician.

- Unless an alternative plan is made by the cardiology unit or an ED consultant, intermediate risk patients being discharged from the ED should be referred to the **Chest Pain Nurse Provided Clinic** via the **ED Referral to Heart Centre** order on Cerner (select chest pain nurse clinic from the drop-down 'choose clinic' menu).

- In some cases, intermediate risk patients being discharged from the ED may be suitable for follow up with a private cardiologist. In such cases, it is recommended that the private specialist be contacted where possible for a verbal handover and an appointment time confirmed prior to the patient being discharged from ED.

- In exceptional cases, and only if approved by an ED consultant or cardiology registrar, intermediate risk patients being discharged from the ED may be suitable to follow up for further testing with their regular GP.

Box 1. Options for further outpatient testing in intermediate risk patients

CT Coronary Angiography (CTCA)	Myocardial Perfusion Scan (MPI)	Stress Echo (exercise or dobutamine)
• Best used in younger patients • B-blocker or ivabradine used to achieve HR < 60. • Iodine contrast required. • Only order after approval by cardiology registrar.	• Often a good choice for patients unable to exercise. • Must withhold beta-blocker and abstain from caffeine 12 hours prior to test. • Call nuclear medicine to arrange (62432).	• Must be able to use cycle/treadmill for exercise stress echo. • Provides additional information regarding cardiac structure and function. • Order "Transthoracic exercise stress echo" on Cerner and call Cardiac Investigations to arrange time.

2. Appendix

2.1 Supplemental use of the HEART score

Prior to the development of hs-Trop-only rapid rule out pathways, the use of clinical risk scores, particularly HEART, was recommended in combination with older ("contemporary") troponin assays to identify low risk patients with a MACE rate <2% that could be safely discharged from the ED.

In the original validation studies, which utilised non-high-sensitivity troponin assays, low risk patients (HEART score of 0—3) had a 1.7% risk of MACE at 6 weeks.

Subsequent studies in which HEART was combined with timed hs-Troponin measurement have showed that risk of MACE in the designated low risk group can be reduced to <1%.

The use of risk scores such as HEART is falling out of use as newer rapid rule-out algorithms that incorporate the concept of Δ troponin have been widely shown to be as safe as algorithms including a risk score, while reducing delays to diagnosis and costs associated with unnecessary testing and extended ED length of stay (LOS).

At Alfred Health, calculation and documentation of the HEART score is nevertheless recommended as being good practice and an additional check point, particularly for junior medical staff prior to discharging low risk patients. It is recommended that patients with a HEART score of 4 or more be discussed with a senior ED registrar or consultant prior to discharge.

The HEART Score

		Points
History	Slightly suspicious	0
	Moderately suspicious	1
	Highly suspicious	2
ECG	Normal	0
	Non-specific repolarisation disturbance	1
	Significant ST deviation or new TW inversion	2
Age	<45	0
	45—64	1
	≥65	2
Risk Factors	No known risk factors	0
	1—2 risk factors	1
	≥3 risk factors or known atherosclerotic disease	2
Troponin (initial)	Not elevated	0
	1—3 x upper limit reference range	1
	≥3 x upper limit reference range	2

Score 0—3 = Low risk
Score 4—6 = Intermediate risk
Score 7—10 = High risk

Further guidance on the use of the HEART Score

History

The designations *slightly, moderately* and *highly suspicious* are ultimately at the discretion of the assessing clinician, but the following approach is recommended*:

High risk historical features	Chest pain PLUS diaphoresis Chest pain PLUS vomiting Chest pain that is exacerbated by exertion or strong emotion Chest pain that radiates to either or both arms Symptom described by the patient as feeling similar to previous MI
Low risk historical features	Very brief duration of pain (<5 minutes) Pain described as sharp Well localised pain

Presence of 2 or more high risk features = highly suspicious
Presence of 1 high risk feature with some low-risk features = moderately suspicious
No high risk features present +/- presence of low-risk features = slightly suspicious

*Historical features listed above represent a combination of those used in the protocols for major HEART Score validation studies[3] and those with evidence to support a degree of predictive value.

ECG

2 points	Potentially ischaemic ST depression New potentially ischaemic TW inversion
1 point	Non-specific TW changes Bundle branch blocks Paced rhythms LV hypertrophy Digoxin effect
0 points	Completely normal ECG

Risk Factors

Any of:
- Smoking (current or recent)
- Diabetes mellitus
- Hypertension (diagnosed and/or treated)
- Hypercholesterolaemia
- Family history of premature CAD (<55 years in 1st degree relative)
- Obesity (BMI ≥30)

"Known atherosclerotic disease" means any of:
- Known coronary artery disease (CAD)
- Previous stroke
- Known peripheral arterial disease

Key related documents

Key aligned policy
- Comprehensive Care Policy

Key legislation, acts & standards:
- Charter of Human Rights and Responsibilities Act 2006 (Vic)[1]

Other relevant documents:
- CODE STEMI
- Heart Rate Lowering Pre-CT Coronary Angiogram

PowerPlans/POCS/QRGs
- Nil

References

1. Brieger D, et al. National Heart Foundation of Australia & Cardiac Society of Australia and New Zealand: Comprehensive Australian Clinical Guideline for Diagnosing and Managing Acute Coronary Syndromes. Heart, Lung and Circulation. 2025 April;34(4): 309-397
2. Mahler SA, et al. The HEART Pathway randomized trial: identifying emergency department patients with acute chest pain for early discharge. Circ Cardiovasc Qual Outcomes. 2015 Mar;8(2):195-203.
3. Byrne RA, et al. 2023 ESC Guidelines for the Management of Acute Coronary Syndromes. European Heart Journal (2023) 44: 3720–3826

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First Issued: 02/10/2019		
Current Version: Approved by: Carl Luckhoff	Title: Director Alfred Emergency & Trauma	Date: 12/08/2025
Endorsed by: Operations Comprehensive Care Committee		Date: 23/09/2025
Disclaimer: This guideline has been developed within the context of Alfred Health service delivery. Alfred Health shall not be responsible for the use of any information contained in this document by another organisation outside of Alfred Health.		

[1] REMINDER: Charter of Human Rights and Responsibilities Act 2006 – All those involved in decisions based on this guideline have an obligation to ensure that all decisions and actions are compatible with relevant human rights.